Program: **Diploma Cyber Forensics and Information Security**	
Course Code: **5287**	Course Title: **Network Security Lab**
Semester: **5**	Credits: **1.5**
Course Category: **Program Core Course**	
Periods / Week: **4 (L:4 T:0 P:0)**	Periods / Semester: **45**

Course Objectives:

- To Identify attacks and efficiently block the attacks and to Develop firewall based solutions against security threats
- To employ access control techniques to the existing computer platforms.
- Installation of relevant software to Demonstrate Virtual box, port scanning, Finding active machines and version of remote OS.
- To demonstrate active and passive fingerprinting, sniffing the router traffic, use of dumpsec.
- To Demonstrate IDS, Rootkits, Open ssl command, setup and monitoring honeypot and to Study a security related problem and recommend solutions.

Course Prerequisites:

Topic /Description	Course Code	Course Title	Semester
Knowledge of Computer Network Softwares and Components.		Computer Network Engineering Lab	IV
Basic Knowledge of problem solving		Problem Solving and Programming	II

Course Outcomes:

On completion of the course, students will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Identify attacks and efficiently block the attacks and to develop firewall based solutions against security threats.	9	Applying
CO2	Understand the use of access control techniques.	11	Applying

CO3	Install and use relevant network softwares to identify the various properties of a network and demonstrate active and passive fingerprinting and sniffing the router traffic.	8	Applying
CO4	Demonstrate IDS, Rootkits, Open ssl command, setup and monitor honeypot and to Study a security related problem and recommend solutions.	14	Applying
	Lab Examination	3	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3		3	3			2
CO2	3		3	3			2
CO3	3		3	3			2
CO4	3		3	3			2

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Name of Experiment	Duration (Hours)	Blooms Taxonomy Level
CO1	**Identify attacks and efficiently block the attacks and to develop firewall based solutions against security threats.**		
M1.01	Network and Service Configuration : Get familiar with VNetLab platform; figure out the topology of given virtual network using network tools; configure routing tables;	3	Applying
M1.02	Setup dynamic services such as Web, SSH and NFS server.	3	Applying
M1.03	Firewall configuration using iptables: Develop a set of ip table rules according to the given specifications.	3	Applying
CO2	**Understand the use of access control techniques and To install and use relevant network softwares to identify the various properties of a network.**		

M2.01	Implement access control lists	3	Applying
M2.02	Learn to install Wine/Virtual Box/ or any other equivalent s/w on the host OS. Perform an experiment for Port Scanning with nmap, superscan or any other equivalent software.	5	Applying
M2.03	1. Using nmap i. Find Open ports on a system ii. Find machines which are active iii. Find the version of remote OS on other systems iv. Find the version of s/w installed on other system (using nmap or any other software)	3	Applying
	Lab examination	1.5	
CO3	**Demonstrate active and passive fingerprinting and sniffing the router traffic.**		
MO1	Perform an experiment on Active and Passive finger printing using XProbe2 and nmap	4	Applying
MO2	Perform an experiment to demonstrate how to sniff for router traffic by using the tool Cain and Abel / wireshark / tcpdump.	4	Applying
CO4	**Demonstrate IDS, Rootkits, Open ssl command, setup and monitor honeypot and to Study a security related problem and recommend solutions.**		
MO1	Demonstrate Intrusion Detection System (IDS) using any tool eg. Snort or any other s/w	2	Applying
MO2	Install RootKits and study the features.	2	Applying
MO3	Generate minimum 10 passwords of length 12 characters using open ssl command	2	Applying
MO4	Setup a honey pot and monitor the honey pot on network.	2	Applying
	Open ended projects**	6	
	Lab examination	1.5	

****Open-ended Experiments:**

(Not for End Semester Examination but compulsory to be included in Continuous Internal Evaluation. Students can do open-ended experiments as a group of 2-3. There is no duplication in experiments between groups.)

Perform wireless audit on an access point or a router and decrypt WEP and WPA (Net

Stumbler)

Demonstrate intrusion detection system (ids) using any tool (snort or any other s/w)

Text / Reference:

T/R	Book Title / Author
T1	Build Your Own Security Lab: A field guide for network Testing, Michael Gregg, Wiley India edition, ISBN: 9788126516919.
T2	Matt Bishop, Introduction to Computer Security, Addison-Wesley, November 5, 2004, ISBN 0-321-24744-2.